

OpenADMET Activity Data Deposition Guidelines

These guidelines set standards for depositing activity data in `catalogs/activities/`. Review these requirements before submitting a PR or updating a dataset. We want any user loading our Parquet files to be able to interpret the data on their own. They should know its origin, derivation, and meaning. They should not need the original notebook, filename, or repository.

Terminology

Term	Meaning	Example
Collection	A directory grouping all versions of one endpoint	ChEMBL_pChEMBL_IC50/
Deposition	One source-version snapshot of a collection	ChEMBL37_IC50/
Dataset	One parquet file plus its catalog entry	ChEMBL_IC50_CYP3A4_CHEMBL340_aggregated.parquet

1. Directory layout and naming

None

```
catalogs/activities/  
└─ <SOURCE>_<ENDPOINT>/          # collection  
    └─ README.md                  # required, see §5  
    └─ curate_data_<ENDPOINT>.ipynb # required, see §6  
    └─ CATALOG_<SOURCE><VER>_<ENDPOINT>.yaml # one per deposition  
    └─ <SOURCE><VER>_<ENDPOINT>/      # deposition (flat)  
        └─ <SOURCE>_<ENDPOINT>_<SUBSET>_raw.parquet  
        └─ <SOURCE>_<ENDPOINT>_<SUBSET>_aggregated.parquet
```

- Spell each source one way, with capitalization consistent with the official source name. E.g. use 'ChEMBL'. Do not use 'CHEMBL' or 'chembl'. Use this same spelling in the directory and filenames.
- Keep deposition directories flat. Use filename suffixes for variants like `scaled` or `trimmed`. Do not use subdirectories.
- Write files from one source version into each deposition directory.
- Link one catalog entry to every file.

2. Required columns - aggregated files

An aggregated file is the model-ready product and is the file type most consumers load. Row labels must be descriptive and contain the filename information,, because the filename and directory path are discarded the moment the file is read into a `DataFrame`. Below are common example column name components and their definitions.

Column	Type	Description	Example
OPENADMET_CANONICAL_SMILES	string	RDKit-canonical SMILES. The deduplication key.	<chem>O=C=NC1=CC=C(C1)C=C1</chem>
OPENADMET_INCHIKEY	string	InChIKey derived from <code>OPENADMET_CANONICAL_SMILES</code> , not from the source SMILES.	ADAKRBAJFHTIEW-UHFFFAOYSA-N
OPENADMET_TARGET	string	Target, tissue, or system measured.	CYP3A4, HLM, Caco2
OPENADMET_TARGET_ID	string	Source identifier for the target; null where the endpoint has no target.	CHEMBL340
OPENADMET_ENDPOINT	string	What was measured.	IC50, LogD, CL, Papp
OPENADMET_UNITS	string	Units of the value columns, verbatim from the source, or the derived unit.	10 ⁻⁶ cm/s, mL.min ⁻¹ .kg ⁻¹ , log10(cm/s)
OPENADMET_SOURCE	string	Originating database.	ChEMBL, PubChem
OPENADMET_SOURCE_VERSION	string	Source release, or the assay accession.	37, AID_743085

Column	Type	Description	Example
<value>_mean	double	Mean across source measurements for the compound.	5.297074
<value>_median	double	Median across source measurements.	5.31
<value>_std	double	Standard deviation. Null when n_records = 1 - this is expected, not missing data.	1.187785
n_records	int64	Number of source measurements behind the row.	3

Follow these guidelines when choosing column names:

- Use 'n_records' instead of 'assay_id_count'. This clarifies if the count reflects assays, documents, or measurements.
- Remove columns containing only null values for the endpoint.
- Include the transform in the name of derived columns. Use names like 'LogPapp' or 'LogUnbound'. Document the exact basis, such as the multiplier, clipping floor, or log base, in the collection README.
- Create separate datasets for trimmed data or outlier removal. Record the bounds in the README. Do not overwrite original values in the primary dataset. Keep the units column. It is not redundant. Users might view a DataFrame in a separate notebook without access to the original README.

3. Required columns - raw files

- **Preserve every field needed to reproduce the aggregation from the raw file.** For ChEMBL that means at minimum standard_value, standard_units, standard_relation, standard_type, and activity_comment.
- **standard_relation and standard_units are not optional.** Without standard_relation, a censored >10 μM measurement is indistinguishable from an exact one, and any aggregate computed over the mixture is wrong. Without standard_units, the value column's meaning cannot be verified against the source.

- **Preserve document provenance** - `doc_id`, `doc_doi`, `doc_pubmed_id`, `doc_year` - so a consumer can trace a measurement to its publication.
- **Carry the same `OPENADMET_*` identity columns** as the aggregated file (§2), so raw and aggregated can be joined and each is independently interpretable.
- **If a source field is dropped deliberately, say so and say why in the README.**

4. Dtypes

- **A column's type must not depend on whether its values happened to be present.** Pandas and Arrow will infer `null` for an all-empty column and `string` for the same column in the next file. Cast to the declared dtype before `to_parquet` rather than relying on inference. An all-null string column is written as `string`.
- **One dtype per column name across all of `catalogs/activities/`.** The same column name never means two different types in two files.
- **Identifiers are strings.** `CHEMBL340`, PubChem AIDs, and target IDs are labels, not integers.
- Booleans are `bool`, not `0/1` or `"True"/"False"`.

5. Collection `README.md`

Every collection has a `README.md` recording, for each deposition:

- **Source, version, and retrieval date.**
- **Selection criteria, in words.** Which assay types, [BioAssay Ontology \(BAO\)](#) terms, units, organisms, and confidence scores were required, and what was excluded. A reader should be able to reconstruct the query intent without reading the notebook.
- **Aggregation rule.** What rows are grouped on, which statistic is reported, and how censored (`>`, `<`) and qualified records are treated — included, excluded, or treated as exact.
- **Units for every value column**, and the exact transform for every derived column.
- **Manual intervention.** Trimming bounds, hand-edited records, anything not produced by the notebook end to end.
- **Source license and required attribution.**
- **A column table** in the format of `pubchem_tox21/final/SCHEMA.md` - column name, data type, description, example. That file is the model to copy.

6. Curation notebook

one notebook per collection should be committed alongside the data, that is able to regenerate the deposition from scratch. It should include:

- **Parameters in one named cell at the top** - source version, targets, bucket, output prefix. Not scattered through the notebook where a re-run can pick up a stale value.

- **Runnable start to finish**, in order, without manual cell selection.
- **Build the catalog fresh from the URIs produced by this run.** Never load a previous version's catalog and add entries to it. The resulting file carries the previous version's URLs for every entry you didn't touch, and it will be published under the new version's name.
- **Entry keys from each catalog assigned exactly once.** If two variants are written to the same key — say a scaled and an unscaled dataset both assigned to `f"{target}_aggregated"` — one silently overwrites the other.
- **Upload paths match the local paths.** If the notebook writes to `scaled/x.parquet` it uploads to `<prefix>/scaled/x.parquet`, not to `<prefix>/x.parquet`.
- **No credentials and no credential shell-outs** (`aws configure get ...`) in a committed notebook. Read from the environment or from the toolkit's settings.
- **Clean before committing.** Remove exploratory leftovers, help cells, unused variables, and dead cells.

7. Catalog YAML

- **Populate metadata on every entry.** At minimum: `description`, `units`, `endpoint`, `target`, and `source_version`. Empty `metadata` (`{}`) is not acceptable: the catalog is the only thing a remote consumer sees before downloading, and it is where `intake` surfaces dataset descriptions.
- **Every URL in a catalog resolves under that catalog's own deposition prefix.** For instance a `CATALOG_ChEMBL37_*.yaml` cannot contain a `ChEMBL35_*` URL.
- **Entry names are `<subset>_<raw|aggregated>` and stable across versions** of the same collection, so a consumer can move between source versions by swapping the catalog file alone.
- **No unreferenced data blocks.** If a `data:` block has no `entries:` key pointing at it, either expose it or remove it.
- **Committed catalogs stay in sync with the S3 objects they name.** Re-push the catalog whenever the data changes.

8. What not to commit

- **Large, re-downloadable source files.** Anything over ~10 MB that can be fetched from the source on demand belongs in the README as an accession plus retrieval date, not in git history.
- `.ipynb_checkpoints/`, `__pycache__/`, `Untitled*.ipynb`, and scratch notebooks. `.gitignore` covers checkpoints; check `git status` before committing.
- **Notebook outputs containing credentials, tokens, or absolute local paths.**

9. PR checklist

Copy into the PR description:

None

- [] Source spelled consistently; deposition directory is flat
- [] One source version per deposition directory
- [] Every parquet file has a catalog entry; every entry resolves to a file
- [] Aggregated files carry all OPENADMET_* identity columns (target, endpoint, units, source, version)
- [] n_records present; no structurally all-null columns shipped
- [] Raw files retain standard_relation, standard_units, standard_type, activity_comment, doc_* provenance
- [] Dtypes explicit and consistent across files; identifiers are strings
- [] Collection README updated: version, retrieval date, selection criteria, aggregation rule, units, licence, column table
- [] Curation notebook re-runnable, parameters in one cell, no credentials
- [] Catalog built fresh this run; no entry key assigned twice; no cross-version URLs
- [] Catalog metadata populated on every entry
- [] No checkpoints, scratch notebooks, or large re-downloadable source files